Ajmain Muhtady Joarder

Email | LinkedIn: amjoarder | GitHub: amjoarder | Google Scholar

EDUCATION

M.Sc. Engg. in Computer Science and Engineering

Jashore University of Science and Technology, Bangladesh | February 2027 (expected)

B.Sc. Engg. in Computer Science and Engineering

Jashore University of Science and Technology, Bangladesh | 2025 | CGPA 3.66

Ranked 4th overall in the department. GPA of 3.96 in both the 7th and final semesters; average GPA of 3.85 over the last five semesters.

Higher Secondary Certificate (HSC)

Govt. M. M. City College, Khulna | Science | GPA 5.00 | 2018

General Category Scholarship; ranked 142nd in the Board of Intermediate and Secondary Education, Jessore.

Secondary School Certificate (SSC)

Khulna Zilla School, Khulna | Science | GPA 5.00 | 2016

Talent Category Scholarship; ranked 35th in the Board of Intermediate and Secondary Education, Jessore.

RESEARCH INTERESTS

Computer Vision; Multimodal Learning; Natural Language Processing; Large Language Models; Reliable and Trustworthy AI.

PUBLICATIONS & MANUSCRIPTS

A. M. Joarder, et al. *A Survey on Human Behavior Detection: From Traditional to Multimodal Deep Learning Approaches.* Under revision at *Computers in Human Behavior Reports*, Elsevier.

Survey of classical methods, deep learning, and multimodal approaches, covering vision-language models, datasets, evaluation metrics, and future research directions.

RESEARCH EXPERIENCE

Bengali Language Model Hallucination Research

Master's thesis | Supervisor: Prof. Dr. Md. Alam Hossain | In progress

BengHalluGen: A Naturalistic Hallucination Taxonomy and Benchmark for Native Bengali Language Models (working title).

- Investigating naturally occurring hallucinations and factual reliability in Bengali language models, with multilingual models as comparative baselines.
- Developing a benchmark for factual question answering, summarization, and open-ended generation, alongside a Bengali-focused taxonomy and human annotation framework.
- Planning evaluation of hallucination rates, error categories, and inter-annotator agreement.

Real-time Violent Behavior Detection

Undergraduate thesis | Supervisor: Jubayer Al Mahmud | UGC-funded research project

Real-time Detection of Violent Behavior in Educational Institutions Through Deep Vision-Based Surveillance.

- Implemented a YOLOv11 detection system and a deep learning pipeline to process live CCTV footage in educational environments.
- Created and manually annotated a custom dataset for the surveillance setting.

PROFESSIONAL EXPERIENCE

Teaching Assistant

Department of Computer Science and Engineering, JUST | August 2025 - Present

- Assist faculty with lectures, lab sessions, and assessment across Data Structures, Digital Signal and Image Processing, Computer Security, Computer Programming and Applications, E-Commerce and MIS, ITEE, and BDSET.
- Prepare teaching, examination, lab, and viva materials; evaluate assignments, lab reports, and examinations; invigilate final examinations.

Network Engineer Intern

Bangladesh Research and Education Network (BdREN) Trust | February 2025 - May 2025

- Assisted with network configuration, monitoring, and troubleshooting using routers, switches, firewalls, and servers; collaborated with technical teams to resolve connectivity issues.

Content Creator

Shabash Fakibaj LLC | June 2025 - Present

- Create short Bengali educational videos about scholarships and student opportunities.

TECHNICAL SKILLS

Programming: Python, C++
Deep Learning: Computer Vision, Natural Language Processing, Multimodal Learning, Transformers
Frameworks & Tools: PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn, Git/GitHub, LaTeX

TRAINING AND CERTIFICATIONS

- **Data Science and Machine Learning with Python** - Ostad
- **Machine Learning courses** - Andrew Ng, Coursera: Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommenders, Reinforcement Learning
- **C for Everyone: Programming Fundamentals** - Coursera
- **Fundamentals of Research Methodology** - Scholarship School BD

LEADERSHIP AND SERVICE

Treasurer, CSE Club of JUST | 2023 - 2024
Managed budgeting, events, and webinars.

President, Rotaract Club of Sundarban | 2022 - 2023
Led health camps, toy drives, and food distribution.

Organizer, Mohabbat Ka Sharbat | 2022 and 2024
Ramadan fundraising for food distribution.

REFERENCES

Jubayer Al Mahmud | Lecturer
Computer Science and Engineering, Jashore University of Science and Technology, Jashore-7408
Email: ja.mahmud@just.edu.bd

Nazmul Hossain | Assistant Professor
Computer Science and Engineering, Jashore University of Science and Technology, Jashore-7408
Email: n.hossain@just.edu.bd